

TASK 1: Fake News Detection Dataset

About Dataset

❑ Fake News Detection Dataset

Overview

This dataset is designed for practicing fake news detection using machine learning and natural language processing (NLP) techniques. It includes a rich collection of 20,000 news articles, carefully generated to simulate real-world data scenarios. Each record contains metadata about the article and a label indicating whether the news is real or fake.

The dataset also intentionally includes around 5% missing values in some fields to simulate the challenges of handling incomplete data in real-life projects.

Columns Description

title A short headline summarizing the article (around 6 words).

text The body of the news article (200–300 words on average).

date The publication date of the article, randomly selected over the past 3 years.

source The media source that published the article (e.g., BBC, CNN, Al Jazeera). May contain missing values (~5%).

author The author's full name. Some entries are missing (~5%) to simulate real-world incomplete data.

category The general category of the article (e.g., Politics, Health, Sports, Technology).

label The target label: real or fake news.

Why Use This Dataset?

Fake News Detection Practice: Perfect for binary classification tasks.

NLP Preprocessing: Allows users to practice text cleaning, tokenization, vectorization, etc.

Handling Missing Data: Some fields are incomplete to simulate real-world data challenges.

Feature Engineering: Encourages creating new features from text and metadata.

Balanced Labels: Realistic distribution of real and fake news for fair model training.

Potential Use Cases

Building and evaluating text classification models (e.g., Logistic Regression, Random Forests, XGBoost).

Practicing NLP techniques like TF-IDF, Word2Vec, BERT embeddings.

Performing exploratory data analysis (EDA) on news data.

Developing pipelines for dealing with missing values and feature extraction.

A Note on Data Quality

This dataset has been synthetically generated to closely resemble real news articles.

The diversity in titles, text, sources, and categories ensures that models trained on this dataset can generalize well to unseen, real-world data.

However, since it is synthetic, it should not be used for production models or decision-making without careful validation.

File Info

Filename: fake_news_dataset.csv

Size: 20,000 rows × 7 columns

Missing Data: ~5% missing values in the source and author columns.

TASK 2: STUDENTS_DATASET_FOR_GRADUATION_CLASSES_AT_RP

About Dataset

This project focuses on developing a student performance classification system using unsupervised machine learning techniques. The goal is to categorize students into different performance classes based on their scores across multiple subjects. The classes include "Fail," "Pass Class," "Lower Class," "Second Class," and "First Class." The system uses K-Means clustering to group students with similar performance patterns and then maps these clusters to predefined class labels for interpretability.

The dataset is synthetically generated to simulate student scores in subjects such as Machine Learning, Data Mining, Cyber Security, Software Engineering, Research Methodology, and Blockchain. The scores are generated in such a way that they are perfectly separable into the predefined classes, ensuring clear distinctions between performance levels.